Indian Institute of Science Education and Research, Kolkata

PH3203 - Advanced Quantum Mechanics

Term Paper

Quantum Teleportation

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Chapter 1

Introduction

1.1 Motivation and context

Information in classical physics is encoded in bits that can be measured and transmitted without limitations. However, quantum information is stored in qubits, which can exist in superpositions of classical bits. This property makes quantum information fundamentally different from classical information.

A major challenge in handling quantum information is that it cannot be freely copied or observed. Measurement disturbs a quantum state, hence an unknown state cannot be measured accurately unless a large copy of states is available. The no-cloning theorem prohibits the duplication of an unknown state, hence having copies is also not possible. As a result, classical methods of communication cannot be applied to quantum systems.

So, how can an unknown quantum state be transmitted between distances without sending the system itself? Quantum teleportation addresses this problem by enabling the transfer of a quantum state using shared entanglement and classical communication. It forms a key component of quantum communication and quantum information processing technologies.

1.2 Qubits

A qubit or quantum bit, is the fundamental unit of quantum information. Unlike a classical bit, which can exist only in one of two definite states (0 or 1), a qubit can exist in a linear combination of both states simultaneously. It is represented as a vector in a 2D Hilbert space:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where α, $\beta \in C$ and they satisfy the normalization condition $|\alpha|^2 + |\beta|^2 = 1$. The ability of a qubit to exist in such a superposition of states distinguishes quantum information from classical information. Upon measurement, the qubit collapses to the state $|0\rangle$ with probability $|\alpha|^2$ or to $|1\rangle$ with probability $|\beta|^2$.

1.3 Entanglement and bell states

Quantum entanglement is a phenomenon in which the state of a composite system cannot be described independently of its subsystems. In an entangled state, the properties of individual qubits are linked to the state of the other, regardless of the separation between them.

A simple and important example of entangled states is given by the Bell states, which are maximally entangled two-qubit states. One of the most commonly used Bell states is

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right).$$

In this state, neither qubit has a definite individual state; however, measurements performed on the two qubits are perfectly correlated. For instance, measuring one qubit immediately determines the state of the other. (if one of them is 0, the other one has to be 0; if one of them is 1, the other one has to be 1).

1.4 Fundamental constraints

Quantum information is subject to constraints that distinguish it from classical information. One of the most important results is the no-cloning theorem, which states that it is impossible to create an identical copy of an unknown quantum state. This prevents the duplication of quantum information for transmission or storage.

Another key limitation arises due to measurement. Measuring a quantum state disturbs it, causing its collapse and resulting in a loss of information. This means that unlike classical systems, where information can be read without alteration, quantum information cannot be accessed without affecting the system.

These constraints make the transmission and manipulation of quantum states non-trivial. They also motivate the need for alternative methods, such as quantum teleportation, which allow the transfer of quantum information without violating these principles.

1.5 Resource requirement

Quantum teleportation requires both quantum and classical resources. The successful transfer of a single qubit from one location to another requires one entangled pair (one ebit) shared between the sender and the receiver, along with the transmission of two classical bits of information. That is:

$$1 \text{ ebit} + 2 \text{ bits} \geq 1 \text{ qubit}$$

The entangled pair acts as a quantum channel that links the two parties, while the classical bits carry the outcome of a measurement performed by the sender. Based on this classical information, the receiver can apply an appropriate operation to recover the original quantum state.

Chapter 2

Quantum Teleportation Protocol

2.1 The unknown quantum state

2.1.1 Mathematical setup

The quantum teleportation protocol begins with an unknown qubit state that Alice wants to transmit to Bob. This state may be written as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where α and β are complex coefficients satisfying the normalization condition

$$|\alpha|^2 + |\beta|^2 = 1.$$

This form represents a general single-qubit state. The word "unknown" means that Alice does not necessarily know the values of α and β. However, the protocol does not require her to know them. The aim is not to measure and determine the state, but to transfer the state itself from Alice's system to Bob's system.

2.1.2 Geometric visualisation: Bloch sphere

A useful geometrical way of visualizing a single-qubit state is the Bloch sphere. In this representation, every pure state of a qubit corresponds to a point on the surface of a unit sphere. The north and south poles represent the basis states $|0\rangle$ and $|1\rangle$, while other points represent superpositions of these basis states. The same qubit can therefore be represented using angular parameters on the Bloch sphere as

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle.$$

This shows that a single qubit may contain both amplitude and phase information. If Alice simply measured the qubit in the computational basis, this phase information would be lost. Therefore, direct measurement cannot be used to transmit an unknown quantum state.

2.1.3 Density matrices

If many identically prepared copies of the state are available, one can estimate the state using quantum state tomography. This involves measuring different observables and reconstructing the density matrix of the state. For a single qubit, the density matrix may be written as

$$\rho = |\psi\rangle\langle\psi|$$

It can also be expressed in terms of the Pauli matrices as

$$\rho = \frac{1}{2}\left(I + r_x X + r_y Y + r_z Z\right),$$

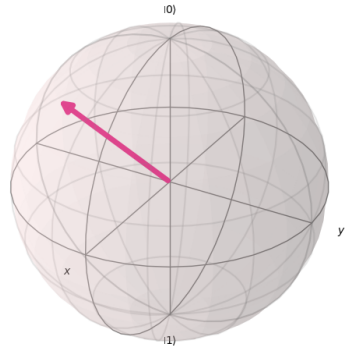

Figure 2.1: Bloch sphere

where I is the identity matrix, and X, Y, and Z are the Pauli matrices given by

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

where (r_x, r_y, r_z) is the Bloch vector.

$$r_x = \langle X \rangle = \mathrm{Tr}(X\rho), \quad r_y = \langle Y \rangle = \mathrm{Tr}(Y\rho), \quad r_z = \langle Z \rangle = \mathrm{Tr}(Z\rho).$$

Thus, r_x, r_y, and r_z are obtained by measuring the qubit along the X, Y, and Z directions, respectively. For example,

$$r_z = P(0) - P(1),$$

where $P(0)$ and $P(1)$ are the probabilities of obtaining the measurement outcomes $|0\rangle$ and $|1\rangle$, respectively.

However, tomography is not the goal of quantum teleportation. Tomography requires many copies of the same state, whereas teleportation is useful even when Alice has only a single unknown qubit. Since the no-cloning theorem prevents Alice from making copies of this state, she cannot determine and then recreate the state at Bob's location. Instead, she uses a shared entangled pair with Bob and sends him two classical bits of information. After Bob applies the appropriate correction operation, his qubit becomes the original state $|\psi\rangle$.

2.2 Initial state

In addition to the initial qubit $|\psi\rangle$, Alice and Bob share an entangled pair of qubits, prepared in a Bell state. A commonly used choice is

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} \left(|00\rangle + |11\rangle \right),$$

where Alice holds one qubit of the pair and Bob holds the other. The total system thus consists of 3 qubits: the unknown state with Alice, Alice's half of the entangled pair and Bob's half of the entangled pair. The combined initial state of the system is given by

$$|\psi\rangle \otimes |\Phi^+\rangle = (\alpha|0\rangle + \beta|1\rangle) \otimes \frac{1}{\sqrt{2}} \left(|00\rangle + |11\rangle \right),$$

we expand this as

$$= \frac{1}{\sqrt{2}} \left(\alpha|0\rangle|00\rangle + \alpha|0\rangle|11\rangle + \beta|1\rangle|00\rangle + \beta|1\rangle|11\rangle \right).$$

Writing the full 3-qubit state:

$$= \frac{1}{\sqrt{2}} \left(\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle \right).$$

2.3 Step 1: Application of CNOT gate

Alice applies a controlled-NOT (CNOT) gate on her 2 qubits. The 1st qubit (the unknown state) acts as the control qubit, and the 2nd qubit (her half of the entangled pair) acts as the target qubit.

A CNOT gate flips the target qubit if the control qubit is in the state $|1\rangle$ and leaves it unchanged if the control qubit is in the state $|0\rangle$. Its action is:

$$|00\rangle \to |00\rangle, \quad |01\rangle \to |01\rangle, \quad |10\rangle \to |11\rangle, \quad |11\rangle \to |10\rangle.$$

Applying this operation to the first 2 qubits of the 3-qubit system, the state transforms as follows:

$$\frac{1}{\sqrt{2}} \left(\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle \right) \xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}} \left(\alpha|000\rangle + \alpha|011\rangle + \beta|110\rangle + \beta|101\rangle \right).$$

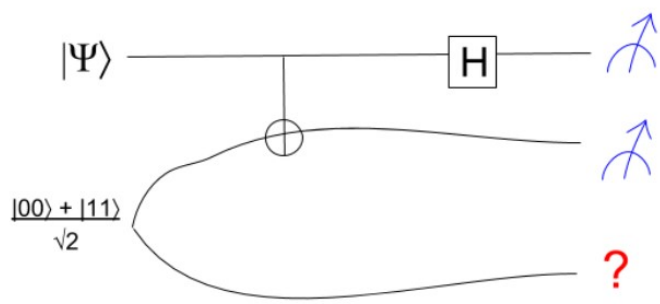

Figure 2.2: Main steps of quantum teleportation.

2.4 Step 2: Application of Hadamard gate

Alice applies a Hadamard gate to her 1st qubit, which is the original unknown qubit. A Hadamard gate transforms the basis states as follows:

$$H|0\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + |1\rangle \right), \quad H|1\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle - |1\rangle \right).$$

This means that the Hadamard gate puts the qubit into a superposition. In this protocol, it is applied only to the 1st qubit of the 3-qubit state.

After the CNOT operation, the state was

$$\frac{1}{\sqrt{2}} \left(\alpha|000\rangle + \alpha|011\rangle + \beta|110\rangle + \beta|101\rangle \right).$$

Applying the Hadamard gate to the 1st qubit gives

$$\frac{1}{2} \left[\alpha \left(|000\rangle + |100\rangle \right) + \alpha \left(|011\rangle + |111\rangle \right) + \beta \left(|010\rangle - |110\rangle \right) + \beta \left(|001\rangle - |101\rangle \right) \right].$$

2.5 Step 3: Measurement of Alice's qubits

Alice measures both of her qubits in the computational basis. These are Alice's original unknown qubit and Alice's half of the Bell pair. Since each measured qubit can give either 0 or 1, Alice can obtain 4 possible outcomes:

$$00, \quad 01, \quad 10, \quad 11.$$

After this measurement, Alice's qubits collapse to one of these 4 outcomes. At the same time, Bob's qubit also collapses into a corresponding state.

The measurement-outcome table is as follows:

Alice's measurement outcome	Bob's qubit state
00	$\alpha\|0\rangle + \beta\|1\rangle$
01	$\alpha\|1\rangle + \beta\|0\rangle$
10	$\alpha\|0\rangle - \beta\|1\rangle$
11	$\alpha\|1\rangle - \beta\|0\rangle$

2.6 Step 4: Classical communication

Alice tells Bob her 2 measurement results through a classical communication channel. These 2 bits tell Bob which of the 4 possible cases occurred after Alice's measurement.

This step is necessary because, without Alice's measurement result, Bob's qubit is only an equal mixture of the 4 possibilities. Once Bob receives the 2 bits, he knows which correction to apply to recover the original state.

2.7 Step 5: Bob's unitary correction

Bob applies a correction depending on the 2 bits received from Alice.

A simple way to write it is:

Alice's bits	Bob's state	Correction
00	$\alpha\|0\rangle + \beta\|1\rangle$	I
01	$\alpha\|1\rangle + \beta\|0\rangle$	X
10	$\alpha\|0\rangle - \beta\|1\rangle$	Z
11	$\alpha\|1\rangle - \beta\|0\rangle$	XZ

If **1st bit** sent by Alice is :

- $1 \rightarrow$ Bob applies the Z transformation,

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

- $0 \rightarrow$ Bob applies the identity operation,

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

If the **2nd bit** sent by Alice is:

- $1 \rightarrow$ Bob applies the X transformation,

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- $0 \rightarrow$ Bob applies the identity operation.

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

2.8 Final state and interpretation

After Bob applies the required correction, his qubit becomes

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle.$$

Thus, Bob has recovered the original unknown state. The qubit has been successfully transmitted using a shared Bell pair and 2 classical bits, without sending the qubit through a quantum channel.

Alice does not keep a copy of the original state. Her measurement is destructive, so the original state on Alice's side is lost. The shared Bell pair is also used up in the process. This is consistent with the no-cloning theorem.

Chapter 3

Extensions and Implications of Quantum Teleportation

3.1 Why the basic protocol is not the end of the story?

In the previous chapter, we discussed the basic quantum teleportation protocol for a single unknown qubit. Alice begins with a qubit in an unknown state, shares an entangled Bell pair with Bob, performs suitable operations and measurements, and then sends 2 classical bits of information to Bob. After Bob applies the appropriate correction, the original quantum state is reconstructed on his side. This completes the teleportation of 1 qubit.

The question is whether the same idea can be applied to more complex quantum systems. For example, what happens if the qubit being teleported is already entangled with another qubit? Can an entangled state itself be teleported?

In particular, the protocol suggests that teleportation can preserve quantum correlations and can be extended to multi-qubit systems. The following sections discuss these consequences. We first consider the teleportation of a qubit that is part of a larger entangled system. Then we examine how this idea leads naturally to multi-qubit teleportation and to the phenomenon of entanglement swapping.

3.2 Teleporting preserves entanglement

In the basic teleportation protocol, we usually describe Alice's qubit as a single unknown state. However, the qubit being teleported does not need to be isolated. It may already be entangled with another qubit or with a larger quantum system.

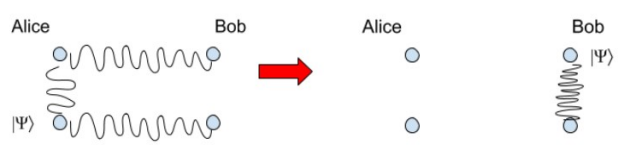

Figure 3.1: Teleportation of a qubit that is initially entangled with another system

Suppose Alice has a qubit, and this qubit is entangled with another qubit. Alice now teleports this qubit to Bob using the usual teleportation protocol. After the protocol is completed, Bob receives the state of qubit. More importantly, Bob's qubit also keeps the entanglement that it originally had.

This means that teleportation transfers not only the local state of a qubit, but also its quantum correlations with other systems. Even though Bob's qubit may never have directly interacted with , it becomes entangled with after the teleportation is completed.

This is a key reason why quantum teleportation is more powerful than simply copying information from one place to another. The original state is not duplicated. Instead, the quantum state, along with its correlations, is reconstructed at Bob's side.

3.3 Extension to multi-qubit states

The idea of teleportation can also be extended from a single qubit to a larger system. Suppose Alice has an arbitrary n-qubit state, written as

$$|\Psi\rangle_{12\ldots N}$$

This state may contain complicated quantum correlations among its qubits. To teleport the full state to Bob, Alice can apply the usual single-qubit teleportation protocol to each qubit one after another. After all qubits have been teleported, the complete state is reconstructed on Bob's side.

The important point is that the protocol does not require the qubits to be independent of one another. Even if the original n-qubit state is entangled, teleporting each qubit separately still transfers the full quantum state, including the correlations between the qubits.

3.4 Entanglement swapping: Creating entanglement without direct interaction

Entanglement swapping is an important consequence of quantum teleportation. It shows that 2 qubits can become entangled even if they have never directly interacted with each other.

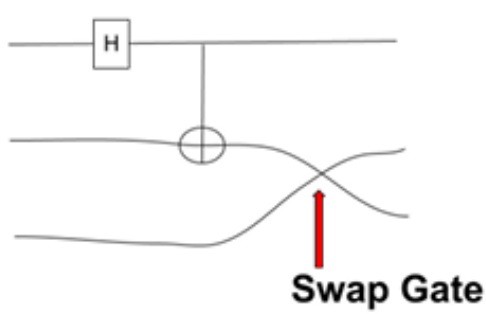

Figure 3.2: Swapping of states

Suppose Alice has 2 qubits, say C and D, that are entangled with each other. Alice also shares 2 Bell pairs with Bob. She can teleport qubit C to Bob using one Bell pair and qubit D to Bob using the other Bell pair. After both teleportations are completed, Bob holds the 2 qubits that originally belonged to Alice. Since teleportation preserves the quantum state and its correlations, Bob's 2 qubits are now entangled with each other. The surprising point is that Bob's 2 qubits never directly interacted with each other. Their entanglement is created on Bob's side because Alice's measurements, classical communication and Bob's correction operations transfer the original entangled state.

3.5 Multipartite entanglement and the GHZ state

Until now, we have mostly discussed entanglement between two qubits, such as a Bell pair. However, entanglement can also involve more than two qubits. This is called multipartite entanglement.

A common example of multipartite entanglement is the GHZ state:

$$|GHZ\rangle = \frac{|000\rangle + |111\rangle}{\sqrt{2}}.$$

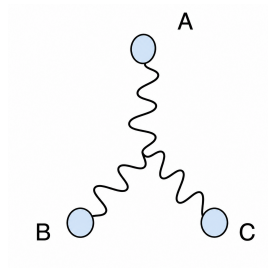

Figure 3.3: GHZ-type three-qubit entanglement shared among Alice, Bob and Charlie.

This is a 3-qubit state shared by 3 parties - Alice, Bob and Charlie. If all 3 qubits are measured in the computational basis, the outcomes are perfectly correlated. They will either all get 0 or they will all get 1. However, the GHZ state is different from ordinary classical correlation. In a classical case, if Alice, Bob and Charlie all hold the same random bit, then even any two of them can check that their bits are correlated. In the GHZ state, the entanglement belongs to the full three-qubit system.

For example, if Charlie's qubit is ignored or removed, Alice and Bob do not share a Bell pair. Their results may still be classically correlated, but they do not have direct pairwise entanglement. This shows that multipartite entanglement can behave differently from two-qubit entanglement.

The GHZ state is an important example because it shows entanglement that is visible only when all three qubits are considered together. It also leads naturally to the idea of monogamy of entanglement, which is discussed in the next section.

3.6 Monogamy of entanglement

Monogamy of entanglement is the idea that entanglement cannot be freely shared among many systems. If one qubit is maximally entangled with another qubit, then it cannot also be maximally entangled with a third qubit.

For example, suppose Alice has a qubit that is maximally entangled with Bob's qubit. In that case, Alice's qubit cannot also be maximally entangled with Charlie's qubit. This is different from classical correlation, where the same information can be shared by many people at once.

The GHZ state gives a useful example of this idea. In the GHZ state, the entanglement belongs to the whole three-qubit system. If Charlie's qubit is ignored, Alice and Bob do not share a Bell pair. They may still have classical correlation, but they do not have maximal pairwise entanglement.

Chapter 4

Summary and conclusion

This term paper explained **quantum teleportation**, a process by which an unknown qubit state can be transferred from Alice to Bob using a shared entangled pair and two classical bits.

A qubit can be written as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where α and β are probability amplitudes. Since an unknown quantum state cannot be copied, Alice cannot simply duplicate the qubit and send it to Bob.

In the teleportation protocol, Alice and Bob first share a Bell pair. Alice performs quantum operations on her qubits, measures them and sends the two-bit result to Bob. Based on this result, Bob applies a correction operation such as I, X, Z, or XZ, where XZ means that Z acts first and then X acts. After the correction, Bob's qubit becomes the original state $|\psi\rangle$.

The protocol does not violate relativity because Bob needs Alice's classical message before he can recover the state. It also does not violate the no-cloning theorem because Alice's original qubit is destroyed during measurement.

The paper also discussed related ideas such as density matrices, tomography, entanglement swapping, GHZ states and monogamy of entanglement. These topics show that entanglement is a powerful but limited quantum resource.

In conclusion, quantum teleportation demonstrates how entanglement and classical communication can transfer quantum information. It is an important foundation for quantum communication, quantum networks, and future quantum computing technologies.

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